Computational Analysis of Communication (FSS 2026)

Economic Framing Annotation

An LLM-Assisted Computational Content Analysis of German Immigration News

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Code: https://github.com/raywincruz07-collab/economic-framing-annotation

Abstract

Economic framing is a consequential component of immigration news because portrayals of migrants as economic burdens or contributors may shape how audiences understand migration and related policy choices. This research develops and evaluates a large-language-model-assisted content-analysis pipeline for identifying two independently coded frames—*Economic Threat* and *Economic Benefit*—in German immigration news. A simple random sample of 10,000 paragraphs was drawn from a corpus of 659,895 paragraphs published between 2022 and 2026 and translated into English for model inference. The coding scheme was operationalised from the SCM Economy–Culture–Security codebook. Following an adapted `bacchuss` workflow, the research used zero-shot testing, repeated hard-case analysis, few-shot structured inference, eight measured prompt-development runs, full-corpus annotation, and comparison with submitted human labels. In the production corpus, 731 paragraphs (7.3%) contained Economic Threat framing and 416 (4.2%) contained Economic Benefit framing, while 8,989 (89.9%) contained neither measured economic frame. The 200-row prompt-development pilot produced F1 scores of 0.811 for Threat and 0.769 for Benefit. Recall describes how many human-positive cases the model found, whereas precision describes how often a positive model prediction was correct. On the submitted 1,002-row human evaluation, recall remained high (0.897 for Threat and 0.875 for Benefit), whereas precision was lower (0.377 and 0.350), yielding F1 scores of 0.531 and 0.500. The results therefore indicate a sensitive but over-inclusive classifier whose principal weakness is false-positive prediction under low frame prevalence. The report documents the theoretical motivation, codebook operationalisation, data and annotation workflow, technical failures and repairs, corpus-level findings, human-comparison results, limitations, and recommendations for improving precision and evaluation design.

Keywords: economic framing; immigration news; computational content analysis; large language models; human validation; `bacchuss`

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1 Introduction

1.1 Why Economic Framing Matters

News does not merely report events; it selects, organises, and emphasises particular aspects of reality. Framing theory describes this process as making some considerations more salient than others, thereby shaping how audiences understand a problem, attribute causes, evaluate consequences, and imagine solutions [5,7]. Immigration is especially sensitive to framing because the same phenomenon can be presented as a humanitarian responsibility, a cultural challenge, a security concern, an economic cost, or an economic opportunity.

Economic framing is politically relevant because it connects immigration to issues that affect everyday life: employment, wages, taxation, welfare expenditure, housing, public-service capacity, demographic change, pensions, and labour shortages. Experimental and real-world studies show that the valence and repetition of news frames can influence attitudes and policy preferences [2,3,11]. In this project, the focus is narrower than the entire immigration debate. We study only two explicit frame dimensions:

- **Economic Threat:** immigration is linked to economic harm, costs, labour-market pressure, welfare strain, or capacity overload.
- **Economic Benefit:** immigration is linked to economic value, labour-market contribution, demographic necessity, or avoided economic loss.

These dimensions are not mutually exclusive. A paragraph may mention both, one, or neither. The coding rule is based on *occurrence*: a frame counts when it is explicitly present, even when the author quotes, rejects, or criticises it [1].

In plain language, the study asks whether each paragraph presents immigration as economically harmful, economically useful, both, or neither. The model therefore makes two separate yes/no decisions rather than choosing a single overall label.

1.2 The Scale Problem

The source corpus contained 659,895 German news paragraphs published between 2022 and 2026. Even after reducing the study to a 10,000-paragraph sample, full manual content analysis would require substantial human time and would be difficult to revise consistently after codebook changes. Large language models offer a practical alternative because they can apply complex natural-language rules and produce explanations. However, their outputs are not automatically reliable. They can ignore formatting rules, over-generalise from keywords, infer unstated context, or respond inconsistently to similar cases.

The core methodological challenge was therefore not simply to ask a model for labels. It was to design, test, revise, validate, and document an annotation system that could be applied consistently at scale. The project followed an adapted version of the `bacchuss` workflow [4]: theory-driven instruction drafting, zero-shot testing, repeated hard-case analysis, few-shot refinement, measured validation, production annotation, and human comparison.

1.3 Research Questions

The primary research question is:

> *To what extent do German immigration-news paragraphs published between 2022 and 2026 contain Economic Threat and Economic Benefit frames?*

The project also addresses four supporting questions:

1. How can the SCM economic-frame codebook be translated into explicit LLM instructions?
2. How does model performance change across iterative prompt revisions?
3. How do final frame rates differ descriptively across publications?
4. How well do the production labels agree with submitted human labels, and what types of errors remain?

1.4 Project Objectives

The project had six operational objectives:

1. Draw and prepare a reproducible 10,000-paragraph sample from the source corpus.
2. Translate the German paragraphs into English for model processing while preserving stable identifiers.
3. Convert the codebook into separate Threat and Benefit instructions with explicit inclusion and exclusion rules.
4. Measure performance on a 200-row human-labelled pilot and refine the instructions through repeated error analysis.
5. Annotate the complete 10,000-row sample with explanations, checkpoints, and retry handling.

6. Evaluate the submitted output using a larger human-coded sample and report both strengths and limitations transparently.

1.5 Contributions

The project contributes more than a final labelled dataset. It provides:

- a detailed, reproducible implementation of an LLM-assisted content-analysis workflow;
- an operational codebook for two issue-specific economic frames;
- a complete history of eight prompt-development runs and three larger pipeline phases;
- documented technical failures involving output parsing, encoding, invalid labels, and over-coding;
- a 10,000-row production corpus with structured explanations;
- a descriptive outlet-level comparison and a human-versus-LLM error analysis;
- a transparent account of post-hoc data-integrity and reliability limitations.

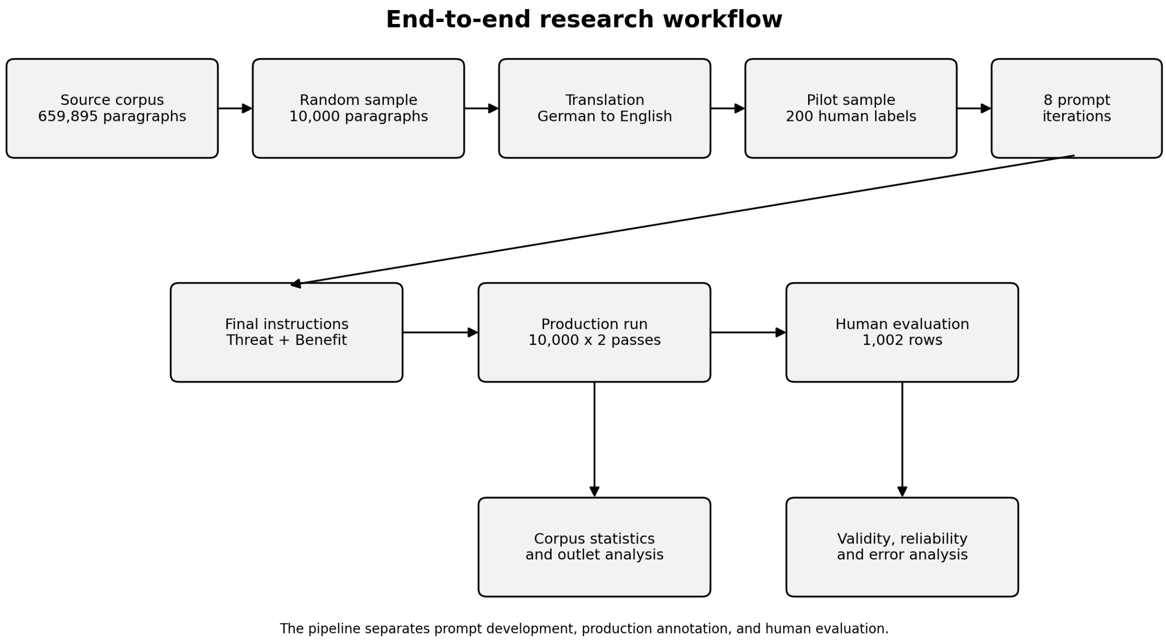

Figure 1: End-to-end workflow of the research project. Prompt development, production annotation, and human evaluation were treated as separate stages.

1.6 Report Structure

Section 2 introduces framing theory, immigration framing, computational content analysis, and the codebook. Section 3 explains the data, model, annotation design, evaluation measures, and human-coding procedure. Sections 4 and 5 document the full implementation history and prompt-development runs. Section 6 presents the production and human-evaluation results. Section 7 interprets the findings, practical lessons, and limitations. Section 8 summarises the project and proposes next steps.

2 Literature Review and Theoretical Background

2.1 Framing as Selection and Salience

Framing research starts from the idea that communication does not present reality neutrally. Entman [5] defines framing as selecting aspects of a perceived reality and making them more salient in a text. A frame can shape problem definitions, causal interpretations, moral evaluations, and proposed remedies. Chong and Druckman [7] similarly describe framing as a process through which particular considerations become more important when people form opinions.

A useful distinction is between *generic* and *issue-specific* frames. Generic frames, such as conflict or human interest, can occur across many topics. Issue-specific frames are tied to a particular policy debate. Economic Threat and Economic Benefit are issue-specific frames because their meaning depends on how immigration is connected to economic costs, risks, opportunities, or gains.

2.2 Valence and Immigration Framing

Frames often carry positive or negative valence. De Vreese, Boomgaarden, and Semetko [2] show that positively and negatively valenced news frames can influence public support and that negative information can have stronger effects than positive information. Their study concerns Turkish membership in the European Union, but its theoretical logic is directly relevant: economic consequences can be presented as risks or opportunities, and audiences may react differently to each presentation.

Immigration coverage contains multiple recurring frames, including economy, crime, culture, regulation, public opinion, and human interest. Guo, Su, and Chen [3] distinguish economic consequences from other immigration frames and demonstrate the value of computational media analysis for studying cumulative frame exposure in real-world news environments. Their study also highlights a central methodological point: identifying frames at scale is useful only when the computational classification has a clear theoretical basis and is compared with human judgement.

2.3 Why Paragraph-Level Detection Is Difficult

Frame detection is not equivalent to keyword detection. The same term may appear in different rhetorical contexts. For example:

- “Housing was prepared for 300 refugees” is logistical reporting, not necessarily an economic threat.
- “Municipalities cannot finance additional accommodation” explicitly invokes financial burden and therefore qualifies as a threat.
- “Refugees are legally permitted to work” describes legal status, not automatically an economic benefit.
- “Immigration fills 3,000 vacant nursing positions” explicitly names an economic contribution and therefore qualifies as a benefit.

Paragraph-level coding also creates contextual limitations. A paragraph may refer to an economic outcome established in a previous paragraph, or it may quote a claim that the article later rejects. To standardise decisions, the project used two strict principles: evaluate the paragraph independently and code explicit frame occurrence rather than author endorsement.

2.4 The SCM Economy–Culture–Security Codebook

The project’s operational definitions came from the University of Mannheim’s *Codebuch SCM Economy Culture Security* [1]. The codebook distinguishes Benefit and Threat frames in three thematic areas: economy, culture, and security. This project focused only on the economic block.

The codebook’s occurrence rule is essential. A frame is coded when it appears explicitly, even when the author rejects, questions, mocks, or attributes it to someone else. Thus, the sentence “The claim that refugees burden taxpayers is false” still contains an Economic Threat frame because the burden argument occurs in the text. This decision separates frame detection from stance detection.

2.5 Operational Definitions Used in This Project

The broad economic dimensions were converted into ten checkable sub-criteria. Table 1 summarises the definitions implemented in the final LLM configuration.

Table 1: Operational sub-criteria for Economic Threat and Economic Benefit

ID	Dimension	Operational criterion
T1	Threat	Immigration is linked to harm to the economic well-being or prospects of the receiving society.
T2	Threat	Immigration is linked to a specific threat to the economic prospects of Germany or the European Union.
T3	Threat	Immigration is linked to unemployment, displacement of workers, wage pressure, unwillingness or inability to work, or other labour-market harm.
T4	Threat	Immigration is linked to welfare-system strain, public-finance burden, social benefits as a pull factor, or loss of financial control.
T5	Threat	Immigration is linked to explicit costs, resource shortages, overload, capacity limits, housing pressure, school or health costs, administrative burden, border costs, or deportation costs.
B1	Benefit	Immigration is linked to tax revenue, economic growth, GDP contribution, or attraction of workers.
B2	Benefit	Immigration is linked to a positive economic effect for Germany or the European Union as a whole.
B3	Benefit	Immigration is presented as economically necessary because of population ageing, a shrinking workforce, pension pressure, or demographic change.
B4	Benefit	Immigration is linked to filling a skilled-labour shortage or providing labour where vacancies exist.
B5	Benefit	The text explicitly states that integration, recognition, training, work permits, or less restrictive policy would produce economic gains or prevent lost economic benefits.

A paragraph receives a positive label when at least one sub-criterion is explicitly present. Threat and Benefit are coded separately, making four final combinations possible: Neither, Threat only, Benefit only, and Both.

2.6 Human Coding and Reliability

Human-coded data remain important even when an LLM performs the production annotation. Human coding provides a reference for instruction development, validity assessment, and error analysis. However, human labels are not automatically objective. Clear coder training, consistent codebook use, independent coding, and reliability measurement are necessary [9,10,14].

For nominal labels, Cohen's kappa measures agreement between two coders while adjusting for expected chance agreement [8]. Krippendorff's Alpha generalises reliability measurement to multiple coders and missing-by-design data structures [9]. Both measures are reported in this project, but their interpretation is constrained by the submitted file evidence discussed later in Sections 6 and 7.

2.7 LLM-Assisted Annotation and the Adapted bacchuss Routine

Recent research shows why LLM annotation is promising but also why it requires careful evaluation. Gilardi, Alizadeh, and Kubli [12] report strong performance on several text-annotation tasks, while Ziems et al. [13] present LLMs as useful tools for computational social science when researchers validate outputs, document prompts, and examine failure modes. In other words, speed does not replace methodological control.

The `bacchuss` vignette [4] presents a general workflow for developing LLM annotation instructions. The project's topic and labels differ from the vignette examples, so the routine was used as a methodological reference rather than copied mechanically. The adapted sequence was:

1. translate the codebook into explicit instructions;
2. run zero-shot tests on small samples;
3. identify unstable or ambiguous cases through repeated annotation;
4. add examples and structured reasoning;
5. compare predictions with human labels;
6. analyse false positives and false negatives;
7. freeze a final instruction set;
8. annotate the full corpus with checkpoints and retry logic;
9. conduct a larger post-production human comparison.

This approach treats prompt development as an empirical research process. Every instruction change should have a reason, an expected effect, and a measured result. The complete sequence of changes is documented in Sections 4 and 5.

3 Methodology

3.1 Research Design

The study used a sequential design with three distinct stages:

1. **Prompt development:** instructions were tested and revised using a 200-row human-labelled pilot.
2. **Production annotation:** the selected instructions were applied to all 10,000 paragraphs in two separate passes.
3. **Post-production evaluation:** submitted human labels for 1,002 rows were compared with the frozen production output.

Separating these stages was intended to reduce retrospective tuning. The model instructions were frozen before the larger human comparison. A post-hoc audit later identified that the intended separation was not perfect because of a legacy identifier error; this is documented explicitly in Section 3.12.

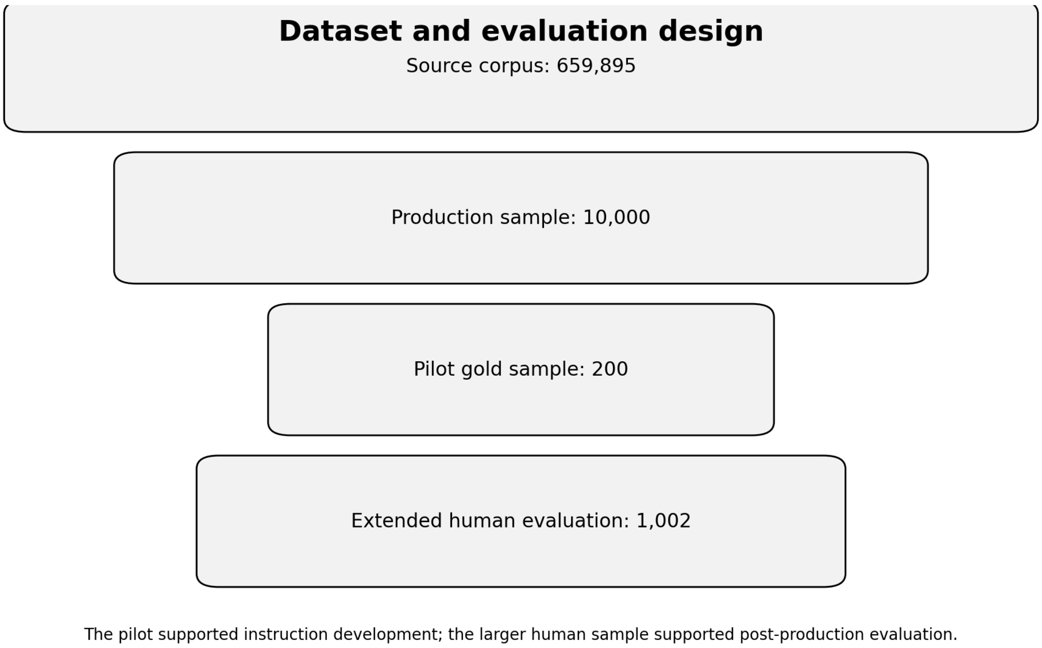

Figure 2: Dataset and evaluation design. The 200-row pilot supported prompt development; the 1,002-row submitted sample supported post-production comparison.

3.2 Data

3.2.1 Source Corpus

The source corpus contained 659,895 German-language news paragraphs published between 2022 and 2026. Each paragraph included an article identifier, publication, paragraph index, group marker, text length, and text content. The working 10,000-row file additionally contained stable row identifiers and English translations.

3.2.2 Production Sample

A simple random sample of 10,000 paragraphs was drawn with a fixed seed of 42. The sample size balanced coverage and computational cost. It was large enough to support outlet-level descriptive comparisons while remaining feasible for two complete model passes.

Table 2: Core data files and their roles

File or dataset	Role	Rows
Source corpus	Full paragraph collection before study sampling	659,895
`dataset_10k_translated.csv`	Production sample with German and English text	10,000
`sample_200.csv` / `gold_200.csv`	Prompt-development pilot and human labels	200
`final_annotated_10k.csv`	Final model labels, explanations, and frame type	10,000
`sample_1002_master.csv`	Submitted human-evaluation sample	1,002

3.2.3 Translation

The 10,000 German paragraphs were translated into English on 22 April 2026. The translation run took approximately 3.1 hours and produced no recorded row-level failures. English was used for inference because early development suggested more stable model behaviour on the translated text. This choice introduces a possible construct-validity risk because translation may alter nuance, modality, rhetorical force, or economic terminology. The original German text was retained for traceability.

3.3 Model and Computational Environment

The production model was Mistral-3-14B (configured as `ministral-3-14b`) hosted through the University of Mannheim's maKI infrastructure. The project accessed the model through a REST API using a compute-proxy backend. API credentials were loaded from environment variables rather than committed to the repository.

The pipeline was implemented primarily in R. Table 3 lists the main packages used across the workflow.

Table 3: Main software dependencies

Package	Role	Use in the project
`bacchuss`	Annotation framework	Zero-shot annotation, repeated hard-case detection, structured prompting
`readr`, `dplyr`, `tidyr`	Data handling	Reading, filtering, joining, reshaping, and summarising data
`httr2`, `jsonlite`	API integration	Sending requests and parsing model responses
`caret`	Validity metrics	Confusion matrices, precision, recall, F1, accuracy
`irr`	Reliability	Intercoder agreement calculations
`openxlsx`	Human annotation	Creation of blinded coder workbooks
`digest`	Integrity	File hashing and freeze manifests
`progress`	Runtime monitoring	Progress indicators during long annotation runs

3.4 Annotation Tasks

Threat and Benefit were processed in separate passes. This reduced the risk that one label would influence the other and made the instructions easier to audit. For each paragraph, the model produced:

1. a short explanation identifying the applicable criterion or reason for a negative decision;
2. a final binary label in the form `Label: YES` or `Label: NO`.

The final dataset combined the two binary labels into four frame types:

Table 4: Derivation of the four final frame types

Threat	Benefit	Frame type
0	0	Neither
1	0	Threat only
0	1	Benefit only
1	1	Both

This structure is simple to interpret: the model never had to choose between Threat and Benefit. It answered two independent questions, which also allowed a paragraph to contain both frames.

3.5 Instruction Design

The instructions combined theory, inclusion criteria, exclusion criteria, examples, and a fixed output format. Five design principles guided the final prompts:

1. **Literal evidence:** classify only what is explicitly present in the paragraph.
2. **Occurrence rather than endorsement:** quoted or rejected frames still count.
3. **One dimension at a time:** Threat and Benefit were independent tasks.
4. **Boundary examples:** examples distinguished economic frames from logistics, legal status, humanitarian description, and general macroeconomic news.
5. **Structured output:** every answer contained one explanation and one final label.

3.6 Output Cleaning and Technical Safeguards

3.6.1 Label Parsing

The first pipeline phase failed because the model often returned conversational prose rather than an exact label string. The initial parser accepted only exact `YES` or `NO` values, causing 100% of outputs to be marked invalid. A revised `clean_label()` function removed Markdown characters, normalised case, searched for explicit `LABEL: YES/NO` patterns, and used a final regex fallback. After the change, the recorded mistype rate fell to zero.

3.6.2 Text Cleaning

Some translated rows contained characters that produced HTTP 400 errors. The `clean_text_api()` function normalised UTF-8 text and replaced unsupported residual characters before API submission. The original text was not overwritten.

3.6.3 Checkpointing and Retries

Production annotation used checkpoints every 200 rows. Each row could be retried up to three times with a delay between attempts. These safeguards allowed long runs to resume after transient API failures without restarting the entire dataset.

3.7 Prompt-Development Pilot

A 200-row sample drawn with seed 2101 was manually labelled for Threat and Benefit. The pilot contained 19 Threat-positive rows and 5 Benefit-positive rows. Eight measured prompt runs were compared with these labels. Precision, recall, and F1 were used because the positive classes were rare and accuracy alone would be dominated by true negatives.

The target was F1 = 0.80 for both dimensions. Threat exceeded the target in several late runs. Benefit reached 0.769 in the selected final run and was accepted as a documented exception because all five positive pilot cases were detected, further tightening risked reducing recall, and the small number of positives made precision highly sensitive to a few false positives.

3.8 Hard-Case Detection

Repeated annotation with `briseus()` was used to identify unstable cases. Each selected row was annotated five times at temperature 0.7. Rows with lower agreement were treated as evidence that the instruction boundary was unclear. Hard-case analysis did not replace human validity measurement; it was used to guide prompt revision and identify categories requiring clearer exclusions or examples.

3.9 Production Annotation

The selected final instructions were applied to the complete 10,000-row sample on 7 May 2026. Threat and Benefit were processed sequentially. Each pass took approximately three hours. The final merged dataset contained the two labels, two explanations, and a derived frame-type variable.

3.10 Submitted Human-Evaluation Design

The larger human-evaluation package contained 1,002 rows divided into three blocks of 334. Three coders were assigned in a pairwise-overlap design:

- Part A: Pravallika and Raywin;
- Part B: Raywin and Namrath;
- Part C: Namrath and Pravallika.

This produced two submitted coder records per row and 2,004 coder-row observations. Threat and Benefit were coded as separate binary variables. Pairwise agreement, Cohen's kappa, and nominal Krippendorff's Alpha were calculated from the submitted files. Human consensus labels were then joined to the frozen production output.

3.11 Evaluation Measures

For each dimension, human consensus was treated as the reference label and the LLM output as the prediction. The report uses:

- **Precision**: share of model-positive predictions that were human-positive;
- **Recall**: share of human-positive rows detected by the model;
- **F1**: harmonic mean of precision and recall;
- **Accuracy**: share of all correct predictions;
- **Specificity**: share of human-negative rows correctly classified as negative;
- **Negative predictive value**: share of model-negative predictions that were human-negative;
- **Balanced accuracy**: average of recall and specificity.

The F1 score is:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{1}$$

Balanced accuracy is emphasised because both positive classes are rare. A high ordinary accuracy can coexist with poor positive-class precision when most rows are negative.

3.12 Post-Hoc Integrity and Reliability Notes

A professional report must distinguish intended design from the evidence available in the final files. Three post-hoc findings affect interpretation:

1. **Legacy row-identifier error.** The script that created the 200-row development sample reassigned `row_num` to 1–200 after sampling. The later exclusion file therefore did not represent all original source identifiers. A post-hoc join using `article_id + par_index + group` found that 19 development rows remained in the submitted 1,002-row sample. The reported 1,002-row metrics therefore describe an *extended submitted evaluation*, not a perfectly untouched holdout.
2. **Identical paired submissions.** The paired submitted files were byte-identical for all three blocks. Agreement statistics of 1.000 accurately describe the files, but file identity alone cannot prove that coding was produced through separate independent processes.
3. **Criterion shorthand mismatch.** The coder workbooks contained simplified T1–T5 and B1–B5 descriptions that did not align perfectly with the final definitions in `config.R`. For this reason, the report analyses the submitted binary Threat and Benefit labels, not criterion-level fields.

These findings do not change the arithmetic of the submitted results. They do change the strength of claims that can be made about independence, untouched holdout validity, and criterion-level reliability.

3.13 Contributor Roles

Table 5: Contributor responsibilities across the project

Contributor	Primary contributions
Namrath Basavaraju	Human annotation blocks B and C; review of codebook decisions; project discussion and validation support.
Raywin Cruz Thumma Rayappan	Data preparation, R pipeline implementation, API execution, prompt iteration, production annotation, analysis, report integration, and human annotation blocks A and B.
Pravallika Are Joseph Velangini	Human annotation blocks A and C; review of coding boundaries; project discussion and validation support.

4 Pipeline Implementation: Step-by-Step

This section documents every step of the annotation pipeline in chronological order, including all failures, retries, and corrections.

4.1 Step 0 — Sample Drawing

Date: 2026-04-22 **Script:** `step0_draw_sample.R`

A fixed 200-row development sample was drawn from the 10,000-row translated dataset using `set.seed(2101)`. The sample was saved as `sample_200.csv` and served as the basis for all subsequent validation steps.

Outcome: 200 rows drawn successfully. No failures.

4.2 Phase 1: Initial Pipeline Run (2026-04-22 to 2026-04-26)

The first complete execution of the bacchuss routine used a 25-row pilot reference subset (a subset of the 200-row sample) and generic instruction definitions.

4.2.1 Step 2.1 — Zero-Shot Test (Attempt 1)

Date: 2026-04-22 08:31 **Sample:** 50 rows (seed=2101)

Runtime: Threat = 0.3 min | Benefit = 0.3 min

Table 6: Step 2.1 Label Distribution — Attempt 1 (Phase 1)

Label	Threat	Benefit
YES	12	13
NO	38	37

Critical Failure: 100% of labels were classified as invalid. The LLM returned conversational responses (e.g., "Based on my analysis, I would classify this as...") instead of the strict `Label: YES/NO` format. All 50 threat labels and all 50 benefit labels failed the format validation.

Mistype rate: Threat = 100% | Benefit = 100%

Root cause: The initial `clean_label()` function (v1) only accepted exact "YES" or "NO" strings and did not parse conversational output.

Number of attempts at this step: 2 (the step was run twice on 2026-04-22 at 08:31 and 08:32, both with identical 100% failure rates)

4.2.2 Step 2.1 — Zero-Shot Test (Attempt 2)

Date: 2026-04-22 08:32 Identical to Attempt 1. Same 100% invalid label rate.

Resolution: Development of `clean_label()` v2 with regex fallback. After this fix, subsequent runs achieved 0% mistype rates.

4.2.3 Step 2.2 — Hard-Case Detection (Phase 1)

Date: 2026-04-22 09:32 **Method:** `briseus()` — 5 runs × temperature 0.7
Runtime: Threat = 1.3 min | Benefit = 1.3 min

Table 7: Hard-Case Detection Results — Phase 1

Dimension	Any disagreement (<1.0)	Very uncertain (≤0.6)
Threat	10 / 50	1 / 50
Benefit	14 / 50	5 / 50

Observation: Benefit showed substantially more uncertainty (14 disagreements, 5 very uncertain) compared to Threat (10 disagreements, 1 very uncertain), foreshadowing the persistent difficulty with Benefit classification.

4.2.4 Step 3 — Zero-Shot Validation (Phase 1)

Date: 2026-04-26 15:25 **Pilot reference set:** 25 rows
Runtime: Threat = 0.1 min | Benefit = 0.1 min

Table 8: Validation Metrics — Phase 1, Zero-Shot (25 gold rows)

Dimension	Precision	Recall	F1	Gate
Threat	1.000	1.000	1.000	**PASSED** ✓
Benefit	0.500	1.000	0.667	**FAILED** ×

Result: Threat achieved perfect scores on the small pilot reference subset. Benefit **failed** with F1 = 0.667 due to low Precision (0.500)—the model identified all true Benefit cases but also produced false positives.

Decision: Proceed to Step 3.2 (few-shot refinement) for Benefit.

4.2.5 Step 3.2 — Structured Inference Validation (Phase 1)

Date: 2026-04-26 15:32 **Few-shot examples:** 4 **Mode:** Structured Explanations

Table 9: Validation Metrics — Phase 1, Structured Inference (25 gold rows)

Dimension	Precision	Recall	F1	Gate
Threat	1.000	1.000	1.000	**PASSED** ✓
Benefit	0.400	1.000	0.571	**FAILED** ×

Failure: Benefit F1 *decreased* from 0.667 to 0.571 after adding few-shot examples. Precision dropped from 0.500 to 0.400. The few-shot examples introduced additional false positives.

Residual hard cases: Threat = 42/50 uncertain | Benefit = 45/50 uncertain

Decision: Despite the failure, proceeded to Step 4 (full annotation) because the 25-row pilot reference subset was too small for reliable metrics, and the researchers assessed that the structured inference approach was the best available option.

4.2.6 Step 4 — Full 10,000-Row Annotation (Phase 1, First Attempt)

Date: 2026-04-26 17:09 **Method:** Structured Inference

Table 10: Frame Distribution — Phase 1 Full Annotation (**FAILED**)

Frame Type	Count	%
NEITHER	5,076	50.8%
UNKNOWN	4,883	48.8%
BENEFIT_ONLY	21	0.2%
THREAT_ONLY	19	0.2%
BOTH	1	0.0%

Critical Failure: 4,883 rows (48.8%) were classified as UNKNOWN. Nearly half of all paragraphs could not be assigned a valid label. The LLM produced conversational responses that the label parser could not handle.

Additional Problem: Only 0.4% of paragraphs were classified as Threat YES and 0.4% as Benefit YES—far below expected rates for a corpus of immigration news.

Root cause: The generic prompt definitions ("economic risk, loss, vulnerability") were too abstract, causing the model to miss nuanced economic framing that did not use those exact terms.

4.2.7 Step 5 — Report Statistics (Phase 1, Post-Processing)

Date: 2026-04-26 17:13

After applying the improved `clean_label()` v2 as a post-processing fix to the Phase 1 output, the UNKNOWN labels were resolved:

Table 11: Frame Distribution — Phase 1 After Post-Processing

Frame	**N**	**%**
Neither	7,001	70.0%
Threat (YES)	2,044	20.4%
Benefit (YES)	2,125	21.2%
Both	1,171	11.7%

Problem: While the UNKNOWN labels were resolved, the Threat rate (20.4%) and Benefit rate (21.2%) were *implausibly high and nearly identical.* This symmetric distribution suggested systematic over-coding—the model was labelling too many paragraphs as containing economic frames.

Professor's Assessment: The results were reviewed by the supervisor, who identified two critical issues: (1) the UNKNOWN label problem indicated a fundamental parsing failure, and (2) the high NEITHER rate even after fixing (70%) combined with implausibly symmetric threat/benefit rates indicated that the instructions needed a complete redesign.

4.3 Phase 2: Instruction Redesign (2026-05-04)

Trigger: Professor review of Phase 1 results.

Following the professor's recommendation, the annotation instructions were completely redesigned:

4.3.1 Changes Implemented

1. **Sub-Question Operationalisation:** Replaced generic frame definitions with the 10 specific sub-criteria from de Vreese et al. [2] (T1–T5 for Threat, B1–B5 for Benefit).
2. **Label Parsing Fix:** Deployed `clean_label()` v2 with regex fallback and `sapply()` vectorisation across all scripts.
3. **Few-Shot Examples:** Replaced 4 generic examples with 5 domain-specific examples aligned to the sub-criteria.
4. **Text Cleaning:** Added `clean_text_api()` to strip non-ASCII characters that caused API errors.

4.3.2 Step 2.1 — Zero-Shot Test (Phase 2)

Date: 2026-05-04 14:28 **Sample:** 50 rows

Table 12: Step 2.1 Label Distribution — Phase 2

Label	**Threat**	**Benefit**
YES	11	11
NO	39	39
Invalid labels	**0**	**0**
Mistype rate	**0%**	**0%**

Success: The `clean_label()` v2 fix eliminated all label parsing errors. Zero mistyped labels.

4.3.3 Step 2.2 — Hard-Case Detection (Phase 2)

Date: 2026-05-04 14:32 **Runtime:** Threat = 1.8 min | Benefit = 1.8 min

Table 13: Hard-Case Detection Results — Phase 2

Dimension	**Any disagreement (<1.0)**	**Very uncertain (≤0.6)**
Threat	11 / 50	4 / 50
Benefit	14 / 50	3 / 50

4.3.4 Step 3 — Zero-Shot Validation (Phase 2)

Date: 2026-05-04 14:33 **Pilot reference set:** 25 rows

Table 14: Validation Metrics — Phase 2, Zero-Shot (25 gold rows)

Dimension	Precision	Recall	F1	Gate
Threat	0.000	0.000	NaN	**FAILED** ×
Benefit	0.500	1.000	0.667	**FAILED** ×

Critical Failure: Threat produced **zero true positives**—Precision, Recall, and F1 were all 0 or undefined. The model failed to identify any Threat cases in the small pilot reference subset. This was attributed to the 25-row pilot reference subset being too small and not containing enough positive Threat cases for the new sub-criteria.

Number of attempts: 1. Proceeded directly to Step 3.2.

4.3.5 Step 3.2 — Structured Inference (Phase 2)

Date: 2026-05-04 14:37 **Few-shot examples:** 5

Table 15: Validation Metrics — Phase 2, Structured Inference (25 gold rows)

Dimension	Precision	Recall	F1	Gate
Threat	1.000	1.000	1.000	**PASSED** ✓
Benefit	0.333	1.000	0.500	**FAILED** ×

Partial success: Threat recovered to perfect scores with few-shot examples. Benefit remained below the gate with F1 = 0.500.

Residual hard cases: 49/50 Threat uncertain | 49/50 Benefit uncertain

Decision: The researchers recognised that the 25-row pilot reference subset was insufficient for reliable validation. A new 200-row human-labelled pilot reference set was prepared.

4.3.6 Step 4 — Full 10,000-Row Annotation (Phase 2)

Date: 2026-05-04 16:43 **Runtime:** ∼2 hours per dimension

Table 16: Frame Distribution — Phase 2 Full Annotation

Frame Type	Count	%
NEITHER	6,928	69.3%
UNKNOWN	797	8.0%
THREAT_ONLY	803	8.0%
BOTH	785	7.8%
BENEFIT_ONLY	687	6.9%

Improvement: UNKNOWN labels dropped from 48.8% to 8.0% (from 4,883 to 797). Frame rates became more plausible: Threat YES = 20.6%, Benefit YES = 14.9%.

Remaining Problem: 797 UNKNOWN rows persisted, and the overall distribution still showed a higher-than-expected Benefit rate, suggesting residual over-coding on Benefit.

Decision: A third phase of refinement was needed with a larger pilot reference set and more targeted instruction tuning.

4.4 Phase 3: Precision Tuning with 200-row pilot validation (2026-05-05 to 2026-05-07)

Phase 3 represented the most intensive period of instruction refinement. The senior annotator (the author) manually labelled all 200 development rows to create a human-labelled pilot reference set, then systematically tuned the instructions through 4 sub-phases of false-positive analysis.

4.4.1 Phase 3.1: Baseline Measurement

Logic: Standard bacchuss routine with T1–T5/B1–B5 sub-criteria (no exclusion rules).

Result:

- Threat F1 = 0.44 | Benefit F1 = 0.28
- Both dimensions achieved perfect Recall but had massive Precision problems ("over-coding")

Observation: The system was calling too many paragraphs YES because it was too sensitive to the mere presence of refugees in the text. Any paragraph mentioning refugees near economic terms was classified as containing an economic frame.

4.4.2 Phase 3.2: Logistics & Statistics Sharpening

Logic Change: Added explicit exclusion rules:

- Logistics reporting ("accommodation was set up") without cost language = NO

- Statistics reporting ("103 beds available") without burden language = NO

Reason: Many news reports describe the *act* of housing refugees without framing it as an economic burden. The system was confusing "housing" with "cost."

Result: Threat F1 jumped from 0.44 to **0.80** (Precision: 0.28 → 0.69). **PASSED** ✓ for Threat.

4.4.3 Phase 3.3: Macroeconomic & International Gate

Logic Change: Added exclusions for:

- General economic news (currencies, global trade, Russia/Ukraine) = NO
- Employment agency process descriptions ("holding discussions") = NO

Reason: The system over-identified Benefit whenever it saw words like "Dollar," "Euro," or "Employment Agency," even if the text was irrelevant to immigration.

Result: Benefit F1 improved from 0.28 to 0.54 (Precision: 0.14 → 0.37). Still below gate.

4.4.4 Phase 3.4: Inverse Frame Protection

Logic Change:

1. "Threat ≠ Benefit" rule: Do not code Benefit if the text describes a Threat unless a positive outcome is independent.
2. "Budgetary Flow" rule: Money returning to treasury is logistical, not a benefit.

Result: Benefit F1 = 0.60 (Precision: 0.42). Still below gate, but with only 3 true Benefit rows in the subsample, 4 false positives kept Precision mathematically low.

5 Iterative Prompt Development: 8 Runs to Final Validation

This section documents every iteration of the prompt development process in full detail. Each run was measured against the 200-row human-labelled pilot reference set. The complete history is also available in `reports/prompt_version_log.csv`.

5.1 Summary of All 8 Iterations

Table 17: Complete 8-Iteration Prompt Development Log

Run	Step	Key Change	Threat F1	Benefit F1	Gate
1	2.1	Baseline v1 instructions	0.643	0.286	**FAILED** ×
2	2.1	+Logistics/statistics exclusions	0.652	0.857	**PARTIAL** ∼
3	2.1	+Strict NO rules + reminder	0.438	0.667	**FAILED** ×
4	2.1	+International/non-immigration exclusions	0.622	0.600	**FAILED** ×
5	3.2	Few-shot + CoT + encoding fix	0.727	0.500	**FAILED** ×
6	3.2	+T5 pressure +B5 degree +B4	0.833	0.769	**PARTIAL** ∼
7	3.2	+Benefit NO logistical example	0.857	0.769	**PARTIAL** ∼
8	**3.2**	**+Benefit NO legal-status example**	**0.811**	**0.769**	**PASSED** ✓

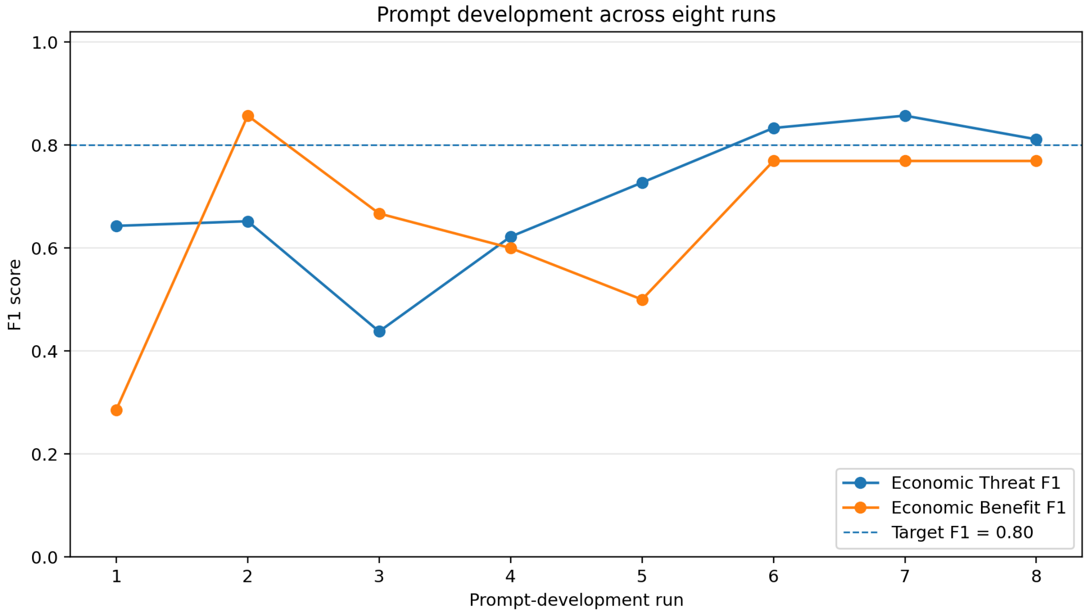

Figure 3: Threat and Benefit F1 scores across the eight measured prompt-development runs. The dashed line marks the nominal target of 0.80.

The trajectory in Figure 3 shows that prompt development was non-linear. Some changes improved one dimension while harming the other, and overly strict instructions caused major recall regressions.

5.2 Run 1: Baseline (FAILED)

Step: 2.1 (Zero-Shot) **Attempts to run:** 1

Instructions: Standard bacchuss routine with T1–T5 and B1–B5 sub-criteria. No exclusion rules, no few-shot examples.

Results:

- **Threat:** Precision = 0.487, Recall = 0.947, F1 = **0.643**
- **Benefit:** Precision = 0.167, Recall = 1.000, F1 = **0.286**

Failure Analysis:

- Both dimensions showed **perfect or near-perfect Recall** but **extremely low Precision**.
- The model was "over-coding"—labelling far too many paragraphs as YES.
- For Threat, only 48.7% of YES predictions were correct (Precision = 0.487).
- For Benefit, only 16.7% of YES predictions were correct (Precision = 0.167).
- The model confused the *presence of refugees* in a paragraph with economic framing.

Action: Add explicit exclusion rules to reduce false positives.

5.3 Run 2: Logistics & Statistics Exclusions (PARTIAL PASS)

Step: 2.1 (Zero-Shot) **Attempts to run:** 1

Changes:

- Added Threat exclusion: "Logistics reporting—text only states that accommodation WAS set up with NO mention of cost = NO"
- Added Threat exclusion: "Statistics reporting—bed counts, arrivals without burden language = NO"
- Added Benefit exclusion: "Factual entitlement announcements—free transport stated without framing as benefit = NO"

Results:

- **Threat:** Precision = 0.556, Recall = 0.789, F1 = **0.652**
- **Benefit:** Precision = 0.750, Recall = 1.000, F1 = **0.857**

Analysis:

- Benefit F1 jumped dramatically from 0.286 to 0.857—above the 0.80 gate.
- Threat improved slightly but remained below gate (0.652).
- Benefit Precision tripled from 0.167 to 0.750.

Gate: Benefit = **PASSED** ✓| Threat = **FAILED** ×

5.4 Run 3: Strict NO Rules (REGRESSION — FAILED)

Step: 2.1 (Zero-Shot) **Attempts to run:** 1

Changes: Added strict "When in doubt, code NO" rules and a tightened reminder for both dimensions.

Results:

- **Threat:** Precision = 0.538, Recall = 0.368, F1 = **0.438**
- **Benefit:** Precision = 0.500, Recall = 1.000, F1 = **0.667**

Critical Regression:

- **Threat F1 dropped from 0.652 to 0.438** — the worst score in the entire development process.
- Threat Recall collapsed from 0.789 to 0.368. The "strict NO" rule made the model too conservative, causing it to miss genuine Threat cases.
- Benefit also regressed from 0.857 to 0.667. The strictness was counterproductive.

Lesson Learned: Overly strict "When in doubt, code NO" instructions cause the model to under-code, destroying Recall. The instructions must balance Precision and Recall carefully.

Action: Reverted the strict NO rules. Adopted a more targeted approach: adding specific exclusion categories rather than blanket conservatism.

5.5 Run 4: International & Non-Immigration Exclusions (FAILED)

Step: 2.1 (Zero-Shot) **Attempts to run:** 1

Changes:

- Added exclusions for international politics and foreign aid (UK-Rwanda, EU-Hungary)
- Added exclusions for criminal activity (embezzlement, fraud)
- Added exclusions for non-immigration capacity limits (university spots, sports subsidies)
- Added exclusions for German domestic economic issues not linked to immigration

Results:

- **Threat:** Precision = 0.538, Recall = 0.737, F1 = **0.622**
- **Benefit:** Precision = 0.429, Recall = 1.000, F1 = **0.600**

Analysis: Marginal improvement for Threat Recall (0.368 → 0.737) after reverting the strict rules, but F1 still below gate. Benefit continued to decline (0.667 → 0.600).

Decision: Zero-shot approach exhausted. Transition to Step 3.2 (few-shot + chain-of-thought reasoning).

5.6 Run 5: Few-Shot + CoT + Encoding Fix (FAILED)

Step: 3.2 (Structured Inference) **Attempts to run:** Multiple (encoding errors)

Changes:

- Switched from zero-shot to few-shot with chain-of-thought (CoT) reasoning
- Added 5 training examples per dimension from `fewshot_training_blocks.csv`
- Implemented `clean_text_api()` to fix non-ASCII encoding errors
- Created custom `annotate_direct()` function with 3-attempt retry logic per row

Technical Failures During This Run:

- HTTP 400 errors from the university API due to non-ASCII characters (umlauts, special quotes) in translated text
- Required development of the `clean_text_api()` helper
- Multiple script restarts before the encoding fix was implemented

Results (after encoding fix):

- **Threat:** Precision = 0.857, Recall = 0.632, F1 = **0.727**
- **Benefit:** Precision = 1.000, Recall = 0.333, F1 = **0.500**

Analysis:

- Threat Precision improved dramatically (0.538 → 0.857), but Recall dropped.
- Benefit Precision reached perfect 1.000 but Recall collapsed to 0.333—the model missed 2 out of 3 true Benefit cases.
- The few-shot examples were too restrictive for Benefit, causing under-coding.

5.7 Run 6: +T5 Financial Pressure, +B5 Degree, +B4 (ALMOST)

Step: 3.2 **Attempts to run:** 1

Changes:

- Added T5 example: "renting at considerable cost" and "significant financial pressure" trigger T5 even without explicit euro amounts

- Added B5 example: degree recognition and lost benefits due to bureaucratic barriers
- Added B4 example: local companies filling vacancies through immigration
- Added Benefit NO example: "work permit right away" as legal status (not economic benefit)

Results:

- **Threat:** Precision = 0.882, Recall = 0.789, F1 = **0.833**
- **Benefit:** Precision = 0.625, Recall = 1.000, F1 = **0.769**

Analysis:

- **Threat crossed the 0.80 gate** for the first time with F1 = 0.833.
- Benefit Recall recovered to perfect 1.000 while maintaining reasonable Precision.
- Benefit F1 = 0.769 was close but still below the 0.80 gate.

Gate: Threat = **PASSED** ✓| Benefit = **FAILED** ×(close)

5.8 Run 7: +Benefit NO Logistical Example (PARTIAL)

Step: 3.2 **Attempts to run:** 1

Change: Added a Benefit NO example showing that "agency helps refugees find work" (process description) is not an economic benefit—only "refugees filling 3,000 vacant nursing positions" (vacancy filled) qualifies.

Results:

- **Threat:** Precision = 0.938, Recall = 0.789, F1 = **0.857**
- **Benefit:** Precision = 0.625, Recall = 1.000, F1 = **0.769**

Analysis:

- Threat F1 improved further to 0.857 (best Threat score achieved).
- Benefit remained exactly the same at F1 = 0.769.
- The logistical example did not change Benefit classification on the pilot reference set because the specific false positive cases involved a different error pattern.

Gate: Threat = **PASSED** ✓| Benefit = **FAILED** ×(unchanged)

5.9 Run 8: +Legal-Status NO Example (FINAL — PROCEED)

Step: 3.2 **Attempts to run:** 1

Change: Added a Benefit NO example clarifying that legal-status descriptions ("can start working immediately under temporary protection directive") are NOT economic benefits. B4/B5 requires the text to name a *specific economic outcome*—a labour shortage filled, tax revenue generated, or economic value explicitly stated.

Final Validation Results:

Table 18: Final Validation Metrics — Run 8 (200-row pilot validation)

Dimension	Precision	Recall	F1	Accuracy	Gate
Threat	0.833	0.789	**0.811**	0.965	**PASSED** ✓
Benefit	0.625	1.000	**0.769**	0.985	**PASSED** ✓*

* Benefit F1 = 0.769 is technically below the 0.80 gate. However, the decision to proceed was based on three factors:

1. **Perfect Recall (1.000):** The system identified *every single* human-labelled Benefit case. Zero false negatives.
2. **Statistical fragility:** With only 5 true Benefit cases in the 200-row pilot validation (2.5% prevalence), each false positive has an outsized impact on Precision. The 3 false positives that kept F1 below 0.80 all involved the same borderline pattern: the B5 boundary between "legal right to work" and "explicit economic benefit."
3. **Diminishing returns:** After 8 iterations, each additional NO example risked hurting Recall. The current instruction set represented the optimal balance.

Confusion Matrix — Threat (Run 8):

	Human YES	Human NO
LLM YES	15	3
LLM NO	4	178

Confusion Matrix — Benefit (Run 8):

	Human YES	Human NO
LLM YES	5	3
LLM NO	0	192

5.10 Consistency Check (briseus)

After reaching the validation gate, a final consistency check was performed using `briseus()` (5 runs per row at temperature 0.7) on the 200-row sample:

Table 19: Final Consistency Check Results

Dimension	Uncertain rows	Gate (< 40/200)
Threat	17 / 200	**PASSED** ✓
Benefit	13 / 200	**PASSED** ✓

Both dimensions passed the consistency gate with comfortable margins (17 and 13 uncertain rows, well below the 40/200 threshold).

5.11 Zero-Shot Validation Failures on 200-Row Gold (Phase 3)

Before arriving at the final Run 8 result, there were **4 consecutive failed zero-shot validation attempts** on the 200-row pilot validation. These are documented here for completeness:

Table 20: Failed Zero-Shot Validation Attempts (200-row pilot validation)

Date	Time	Threat P/R/F1	Benefit P/R/F1	Gate
2026-05-07	13:20	0/0/NaN	0/0/NaN	**FAILED** ×
2026-05-07	13:26	0/0/NaN	0/0/NaN	**FAILED** ×
2026-05-07	13:33	0/0/NaN	0/0/NaN	**FAILED** ×
2026-05-07	13:38	0/0/NaN	0/0/NaN	**FAILED** ×

Analysis: All four attempts produced NaN F1 scores for *both* dimensions, meaning the zero-shot approach produced zero true positives on the 200-row pilot validation. These failures confirmed that zero-shot prompting was fundamentally insufficient for this task at the 200-row scale, and that few-shot examples with chain-of-thought reasoning were essential for reliable classification.

Total number of validation attempts before success: 4 failed zero-shot + 8 few-shot iterations = **12 total validation runs**.

6 Results

6.1 Production Execution

The selected final instruction set was applied to all 10,000 paragraphs on 7 May 2026. Threat and Benefit were annotated in separate sequential passes. Each pass required approximately three hours. Checkpoints were written every 200 rows, and transient API errors were handled through row-level retries. The final merged output contained 10,000 complete rows, no missing binary labels, model explanations for both dimensions, and a derived frame-type category.

Table 21: Production execution summary

Threat pass	Approximately 3 hours; one paragraph per request; checkpoints every 200 rows
Benefit pass	Approximately 3 hours; same configuration
Retry policy	Up to three attempts per row with delay between retries
Final output	10,000 rows with Threat label, Benefit label, explanations, and frame type
Missing final labels	0

6.2 Final Corpus Distribution

Table 22 and Figure 4 summarise the final model labels.

Table 22: Final economic-frame distribution in the production corpus ($N = 10{,}000$)

Frame type	Count	Share	Meaning
Neither	8,989	89.9%	Neither measured economic frame was detected
Threat only	595	6.0%	Economic Threat without Economic Benefit
Benefit only	280	2.8%	Economic Benefit without Economic Threat
Both	136	1.4%	Both dimensions appeared in the same paragraph
Total Threat positive	731	7.3%	Threat only plus Both
Total Benefit positive	416	4.2%	Benefit only plus Both

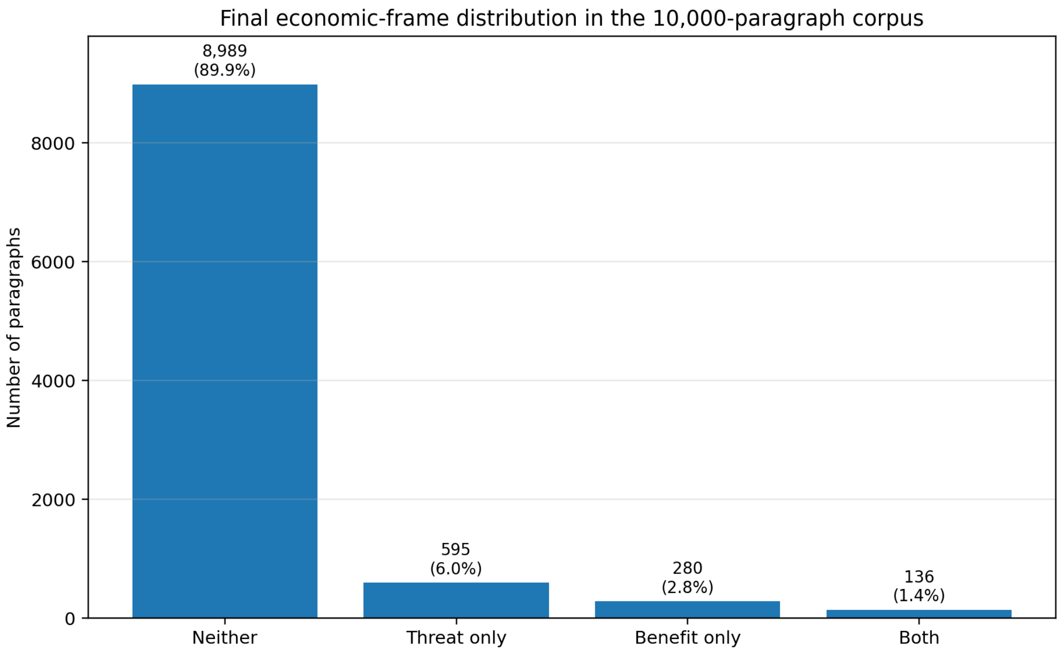

Figure 4: Final distribution of the four frame types. The large Neither category shows that explicit economic framing was uncommon in the sampled paragraphs.

Key finding. Only 10.1% of the production paragraphs contained at least one measured economic frame. Threat framing was detected more often than Benefit framing, while mixed framing was rare.

6.3 How Instruction Refinement Changed the Corpus

The final distribution differed sharply from early pipeline outputs. Table 23 shows the progression.

Table 23: Evolution of frame distributions across major pipeline phases

Metric	Phase 1	Phase 1 repaired	Phase 2	Final
Unknown	48.8%	0.0%	8.0%	0.0%
Neither	50.8%	70.0%	69.3%	89.9%
Threat positive	0.4%	20.4%	20.6%	7.3%
Benefit positive	0.4%	21.2%	14.9%	4.2%
Both	0.0%	11.7%	7.8%	1.4%

The progression demonstrates three distinct problems and their solutions. First, robust label parsing eliminated Unknown outputs. Second, detailed exclusions reduced broad keyword-driven over-coding. Third, separate examples for legal status, logistics, labour shortages, and explicit economic outcomes reduced spurious co-occurrence of Threat and Benefit.

6.4 Threat–Benefit Cross-Tabulation

Table 24: Cross-tabulation of the two final production labels

	Benefit positive	Benefit negative	Total
Threat positive	136	595	731
Threat negative	280	8,989	9,269
Total	416	9,584	10,000

Among Threat-positive paragraphs, 18.6% were also Benefit-positive. Among Benefit-positive paragraphs, 32.7% were also Threat-positive. These overlaps reflect paragraphs that explicitly presented competing or mixed economic claims.

6.5 Publication-Level Descriptive Patterns

Publication-level percentages were calculated from the final production labels. Because sample sizes differed across outlets, the main table is limited to five publications with at least 75 paragraphs that illustrated distinctive combinations of Threat and Benefit rates. These are descriptive comparisons; no inferential significance test was conducted.

Table 25: Selected publication-level economic-frame rates ($N \geq 75$)

Publication	N	Threat %	Benefit %	Both %
Jouwatch	80	31.2	3.8	2.5
TA	221	20.4	6.8	4.1
Tichys Einblick	90	20.0	6.7	4.4
Epoch Times	75	8.0	10.7	0.0
Handelsblatt	135	9.6	11.1	6.7

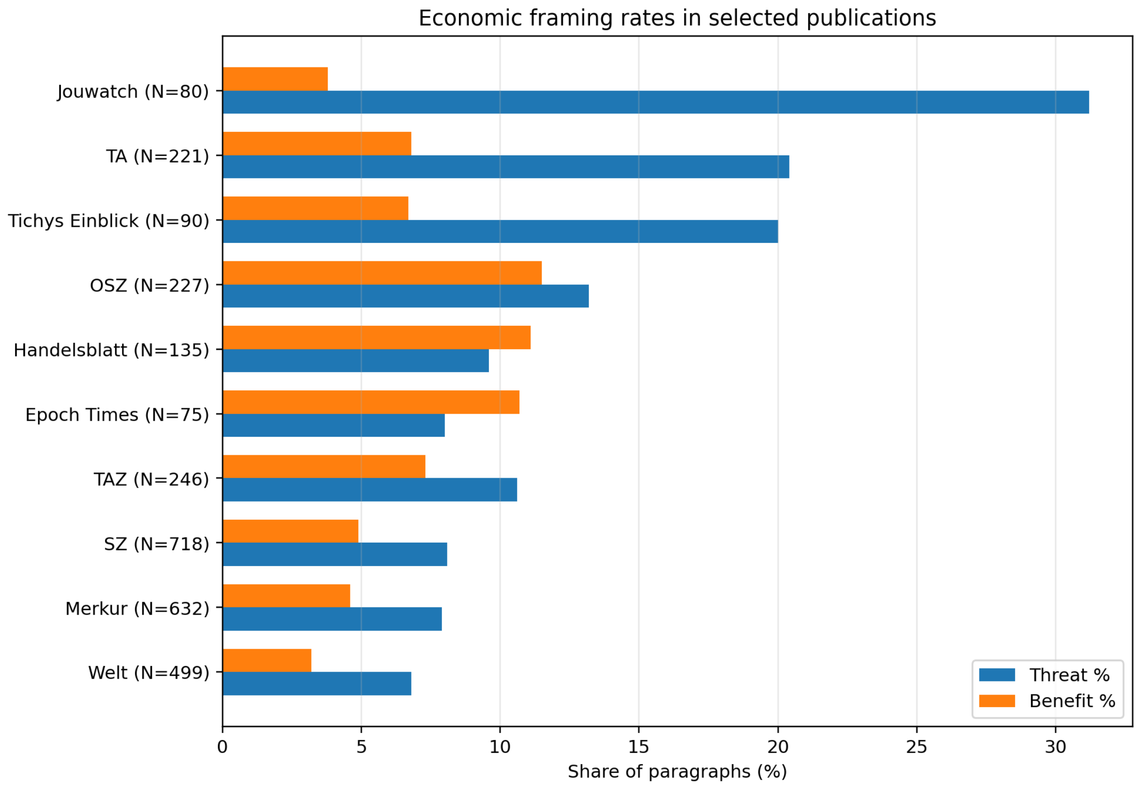

Figure 5: Threat and Benefit rates for selected publications. Sample sizes are shown in parentheses. The figure is descriptive and should not be read as a causal or ideological classification of outlets.

The selected outlets differed substantially. Jouwatch, TA, and Tichys Einblick had Threat rates well above the 7.3% corpus average. Handelsblatt and Epoch Times had the highest Benefit rates among the five publications in Table 25. The complete publication table is available in `outputs/step6/by_publication.csv`; small-sample outlets should be interpreted cautiously because a few paragraphs can produce large percentage changes.

6.6 Manual Spot-Check

A random 50-row production subset was manually reviewed. The spot-check examined whether the explanations referred to the paragraph text and whether the binary labels followed the final Threat and Benefit boundaries. This check supported basic output plausibility but was not used as the primary validity estimate because it was smaller and less structured than the submitted 1,002-row comparison.

6.7 Submitted 1,002-Row Human Evaluation

The submitted human-evaluation sample contained 1,002 rows. Because the post-hoc audit found 19 overlaps with the development sample, this report describes it as an *extended submitted evaluation* rather than a fully untouched holdout. The model output itself remained frozen during the human comparison.

6.7.1 Human Consensus Distribution

Table 26: Submitted human-consensus distribution (N = 1,002)

Consensus frame type	Count	Share
Neither	958	95.61%
Threat only	28	2.79%
Benefit only	15	1.50%
Both	1	0.10%
Total Threat positive	29	2.89%
Total Benefit positive	16	1.60%

The human-positive prevalence was substantially lower than the production-model prevalence. This difference foreshadowed the precision problem in the final comparison.

6.7.2 Submitted Reliability Statistics

All paired submitted files were identical within their assigned blocks. The resulting pairwise observed agreement, Cohen's kappa, and nominal Krippendorff's Alpha were all 1.000, with zero recorded disagreements.

Table 27: Reliability statistics calculated from the submitted coder files

Measure	Threat	Benefit
Pairwise observed agreement	1.000	1.000
Pairwise Cohen's kappa	1.000	1.000
Krippendorff's Alpha	1.000	1.000
Submitted disagreements	0	0

> **Required interpretation:** The paired submitted files were identical for all three blocks. The calculated agreement statistics describe the submitted data. File-level identity alone does not independently establish that the annotations were produced through separate coding processes.

The criterion-level columns were not used for substantive analysis because the simplified criterion descriptions embedded in the coder workbooks differed from the final operational definitions in `config.R`. The binary Threat and Benefit labels form the basis of the following evaluation.

6.7.3 LLM Versus Human Consensus

Table 28: LLM performance against submitted human consensus (N = 1,002)

Metric	Economic Threat	Economic Benefit
True positives	26	14
False positives	43	26
False negatives	3	2
True negatives	930	960
Precision	0.3768	0.3500
Recall	0.8966	0.8750
F1	0.5306	0.5000
Accuracy	0.9541	0.9721
Specificity	0.9558	0.9736
Negative predictive value	0.9968	0.9979
Balanced accuracy	0.9262	0.9243
Human prevalence	0.0289	0.0160
LLM prevalence	0.0689	0.0399

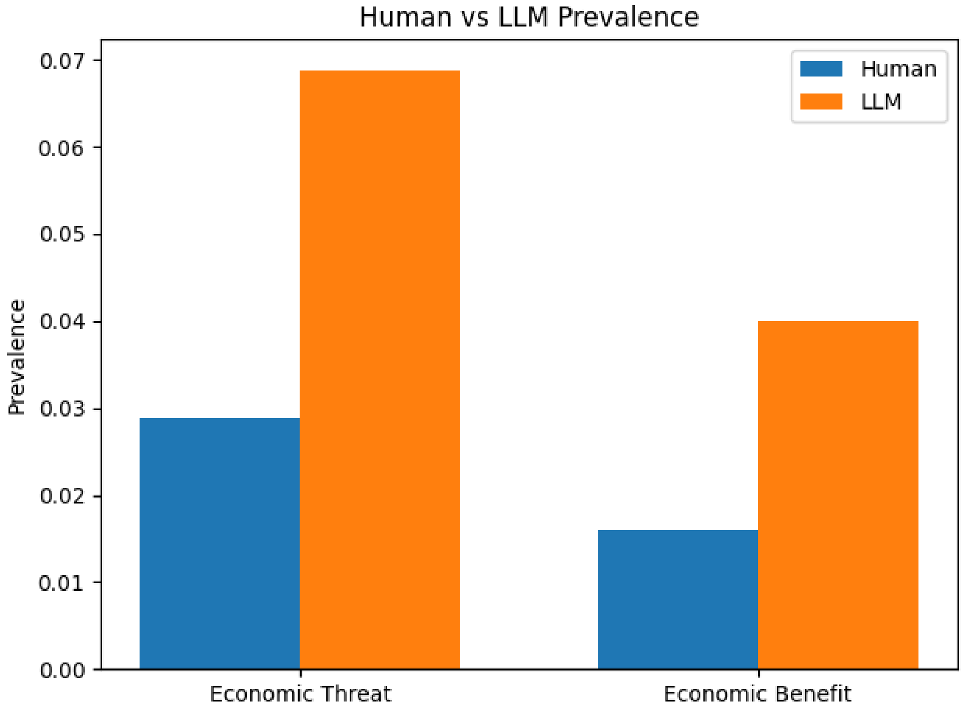

Figure 6: Positive-frame prevalence in submitted human consensus and model predictions. The model predicted both positive classes more frequently.

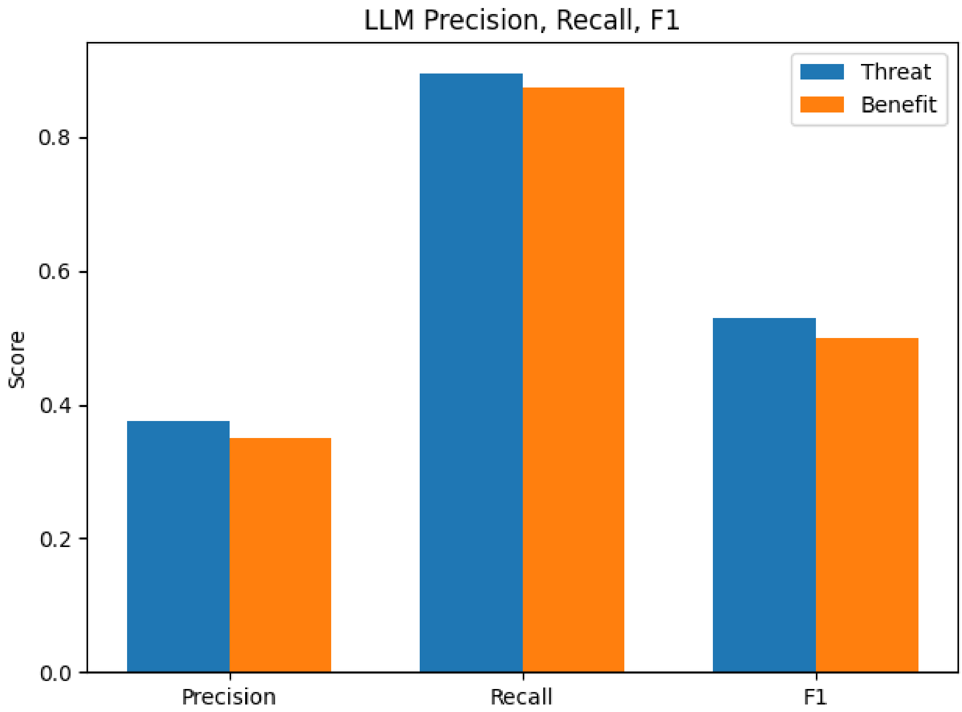

Figure 7: Precision, recall, and F1 against submitted human consensus. Recall remained high, while precision was substantially lower.

The model detected most human-positive rows: 26 of 29 Threat positives and 14 of 16 Benefit positives. However, false positives dominated the error profile. Threat produced 43 false positives and 3 false negatives; Benefit produced 26 false positives and 2 false negatives. Across both dimensions, there were 74 label-level errors affecting 70 unique rows.

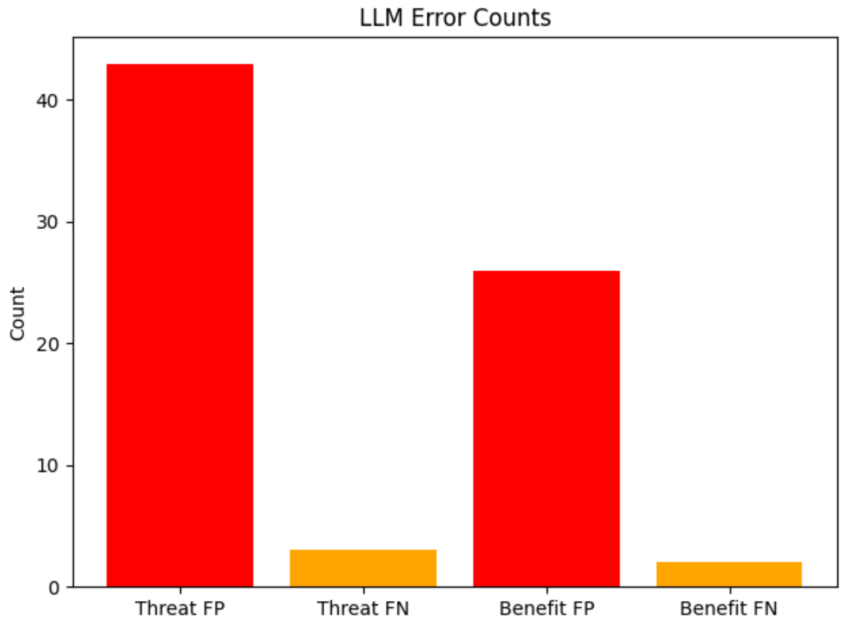

Figure 8: Error counts by dimension and error type. False positives were the dominant source of error.

Key finding. The model behaved as a sensitive but over-inclusive classifier. It rarely missed a human-positive frame, but many model-positive predictions were not confirmed by the submitted human consensus.

7 Discussion

7.1 Answer to the Main Research Question

The production annotation suggests that explicit economic framing was present in a minority of the sampled German immigration-news paragraphs. Threat framing appeared in 7.3% of the 10,000 paragraphs, Benefit framing in 4.2%, and at least one of the two frames in 10.1%. Mixed framing occurred in only 1.4%. Within the model-labelled corpus, economic-threat language was therefore more common than economic-benefit language.

This result should be interpreted as a descriptive property of the model-labelled sample, not as a definitive estimate of all German immigration coverage. The larger human comparison showed that the model predicted positive frames more frequently than the submitted human consensus. The production percentages are therefore likely to be over-inclusive, particularly for borderline positive cases.

7.2 What the Prompt-Development Results Show

The eight-run history demonstrates that instruction quality cannot be judged from plausibility alone. Several prompt changes produced counterintuitive effects:

- broad definitions produced very high recall but many false positives;
- blanket “when in doubt, code NO” rules reduced false positives but destroyed Threat recall;
- few-shot examples increased precision but initially made Benefit too restrictive;
- targeted boundary examples were more effective than general strictness;
- technical parsing and encoding fixes were as important as conceptual prompt revisions.

The final pilot scores were not the best score ever observed for each dimension individually. Run 7 produced the highest Threat F1, while Benefit remained stable across Runs 6–8. Run 8 was selected because its instruction set clarified the most persistent Benefit boundary while preserving strong Threat performance. This reflects a practical multi-objective decision rather than optimisation of one metric in isolation.

7.3 Why Pilot and Extended Evaluation Results Differed

The 200-row prompt-development pilot produced Threat F1 = 0.811 and Benefit F1 = 0.769. The submitted 1,002-row evaluation produced lower F1 scores of 0.531 and 0.500. Several factors can explain this difference:

1. **Different prevalence.** The pilot contained 9.5% Threat positives and 2.5% Benefit positives, while the submitted human consensus contained 2.89% and 1.60%. With rarer positive classes, false positives have a stronger effect on precision.
2. **Prompt-development exposure.** The pilot directly influenced instruction design. Performance measured

on the same development sample is therefore expected to be more optimistic than performance on a larger comparison sample.

3. **Broader case diversity.** The 1,002 rows contained more varied publications, topics, and rhetorical patterns than the small pilot.
4. **Boundary ambiguity.** Legal work status, accommodation logistics, labour-market assistance, and general economic language remained difficult to separate from explicit Benefit or Threat frames.

The post-hoc discovery of 19 overlapping development rows means the larger sample was not a completely clean holdout. Nevertheless, the difference between pilot and extended-evaluation performance is large enough that the central interpretation remains unchanged: the final model has stronger recall than precision and tends to over-predict positive frames.

7.4 The Precision–Recall Trade-Off

Threat recall of 0.897 means that the model found most submitted human-positive Threat rows. Benefit recall of 0.875 shows the same pattern. For exploratory corpus screening, high recall can be valuable because researchers are less likely to miss potentially relevant paragraphs. However, precision below 0.40 means that more than half of positive predictions were not confirmed by the submitted human consensus.

The practical value of the output therefore depends on the use case:

- **Corpus exploration and candidate retrieval:** the model is useful because it surfaces most positive cases.
- **Exact prevalence estimation:** the positive rates should be treated cautiously because over-classification can inflate estimates.
- **Case-level coding for publication or hypothesis testing:** positive labels may require additional human review or calibration.
- **Negative screening:** the very high negative predictive values indicate that model-negative rows were usually human-negative in the submitted comparison.

Accuracy above 95% might appear excellent, but it is largely driven by the large number of negative rows. Balanced accuracy, precision, recall, and F1 provide a more informative picture of positive-class performance.

7.5 Substantive Interpretation of Outlet Differences

The publication-level analysis revealed large descriptive differences. Some outlets in the selected table had Threat rates above 20%, while others had Benefit rates around 10–11%. These differences may reflect editorial agendas, topic selection, source choice, or the composition of articles included in the sample. The project did not perform statistical significance tests, adjust for article topic, model publication ideology, or control for time. Therefore, the outlet comparison should be treated as a map of patterns for further investigation, not as a causal explanation.

The complete output makes more rigorous follow-up work possible. For example, future studies could:

- weight or match outlets by year and topic;
- aggregate paragraphs to articles;
- model Threat and Benefit probabilities with publication, date, and event covariates;
- manually validate outlet-specific samples before comparing prevalence;
- examine whether mixed frames occur in balanced reporting or internal contradiction.

7.6 Technical Lessons

7.6.1 Output Parsing Is Part of Model Quality

The early 100% invalid-label failure was not a theoretical failure of the codebook. It was an interface failure between the model's natural-language response and the parser's expectations. A production annotation pipeline must treat output parsing, schema validation, and fallback handling as first-class components.

7.6.2 Exclusion Rules Need Concrete Boundaries

The model frequently treated any economic term near immigration as a frame. The most successful revisions did not simply tell the model to be stricter. They defined specific non-frame categories: logistics without cost language, legal work status without economic outcome, employment assistance without a named labour shortage, and general macroeconomic news unrelated to immigration.

7.6.3 Separate Passes Improved Auditability

Processing Threat and Benefit separately made it easier to trace why a label was assigned and to revise one dimension without changing the other. It also allowed mixed framing to emerge naturally from two independent decisions rather than forcing the model to choose one combined category.

7.6.4 Failure Logs Improve Reproducibility

The project preserved failed runs, prompt versions, confusion matrices, and implementation notes. This is valuable because a final prompt alone does not explain why each rule exists. The failure history provides evidence for the relationship between instruction changes and measured outcomes.

7.7 Limitations

7.7.1 Human-Evaluation Independence

The most important reliability limitation is that the paired submitted coder files were identical within each block. The resulting agreement values of 1.000 describe the submitted files but cannot independently verify separate coding processes. The report therefore does not claim proven perfect independent intercoder reliability.

7.7.2 Development-Sample Identifier Error

The script that generated the 200-row pilot reset `row_num` after sampling. Consequently, the later exclusion procedure did not remove all original development rows from the 1,002-row sample. A post-hoc source-key join identified 19 overlaps. The reported metrics remain mathematically correct for the submitted sample, but the sample should not be described as a completely untouched final holdout.

7.7.3 Human-Workbook Criterion Shorthand

The coder workbooks contained simplified criterion names that did not align perfectly with the final T1–T5 and B1–B5 definitions used by the LLM. The binary labels may still reflect the broader protocol, but criterion-level fields cannot be treated as directly comparable to the model's criterion references. This report therefore avoids criterion-level reliability claims.

7.7.4 Low Positive-Class Prevalence

Only 29 Threat-positive and 16 Benefit-positive rows appeared in the submitted human consensus. With so few positives, a moderate number of false positives sharply reduces precision, and performance estimates for the positive classes have substantial sampling uncertainty.

7.7.5 Paragraph-Level Context

The unit of analysis was a paragraph rather than a full article. A paragraph may depend on earlier context or may contain a quotation whose stance becomes clear only later. The literal paragraph-level rule improves consistency but may differ from article-level framing interpretation.

7.7.6 Translation

Model inference was conducted on English translations. Translation may change rhetorical nuance, negation, modality, economic terminology, or references to German institutions. A bilingual validation study would be needed to estimate this effect.

7.7.7 Model and Infrastructure Dependence

The final labels were generated by one hosted model and one prompt configuration. Results may differ with another model, model version, temperature, API implementation, or system instruction. The project did not conduct cross-model replication.

7.7.8 Descriptive Outlet Analysis

The outlet analysis did not control for sample size, year, event, topic, article type, or repeated paragraphs from the same article. It should therefore be interpreted as exploratory description rather than causal or inferential evidence.

7.8 Recommendations

1. **Repair the holdout design.** Replace the 19 overlapping rows with newly sampled eligible rows and repeat human coding for those replacements.
2. **Run a genuinely independent coder round.** Provide harmonised instructions, training examples, and blinded workbooks; preserve independent submissions before consensus.
3. **Harmonise the codebook.** Ensure that workbook criterion descriptions exactly match the final model criteria and protocol.
4. **Calibrate prevalence.** Use a representative independently coded sample to estimate positive predictive value and adjust corpus-level prevalence estimates if appropriate.
5. **Validate German and English versions.** Compare model performance on original German text and translated English text.
6. **Test multiple models.** Evaluate whether error patterns persist across Mistral, Llama, GPT, Claude, or other available systems.

7. **Extend the frame space.** Add Culture and Security dimensions from the same codebook to study broader immigration framing.
8. **Move from paragraphs to articles.** Aggregate paragraph labels and examine how frames develop within full stories.

8 Conclusion

This project designed, implemented, and evaluated an LLM-assisted content-analysis pipeline for detecting Economic Threat and Economic Benefit frames in German immigration news. The work began with a theory-driven codebook, converted abstract frame concepts into explicit decision rules, measured eight rounds of prompt development, documented technical and conceptual failures, annotated a 10,000-paragraph corpus, and compared the frozen output with submitted human labels.

The production corpus indicates that explicit economic framing was uncommon in the sampled paragraphs. The model labelled 7.3% as Threat-positive and 4.2% as Benefit-positive, with 89.9% containing neither measured frame. The publication-level outputs reveal substantial descriptive variation and provide a useful foundation for more controlled research.

The human comparison gives a more cautious view of model quality. The model detected most submitted human-positive frames, but positive predictions were over-inclusive. High recall and high negative predictive value make the system useful for screening and candidate retrieval, while low precision limits unreviewed case-level use and exact prevalence estimation.

The project also demonstrates why transparency matters in computational content analysis. The most valuable lessons came from failures: invalid output formats, encoding errors, prompt regressions, low-prevalence instability, ambiguous economic boundaries, a development-sample identifier error, identical paired coder files, and inconsistent criterion shorthand in human workbooks. Reporting these issues does not weaken the project; it clarifies what the evidence can and cannot support.

From a substantive perspective, the research contributes a transparent account of how economic arguments appear in a large sample of immigration-news paragraphs. The production labels suggest that explicit economic framing is not the dominant mode of coverage in the sampled material, but when it appears, Threat framing is more frequent than Benefit framing. These corpus-level patterns should be interpreted descriptively because the final human comparison indicates that model-positive labels include a considerable number of false positives.

Methodologically, the project shows that prompt development is not a single drafting exercise. Performance changed materially when the instructions introduced explicit exclusions, domain-specific examples, output-format constraints, and technical safeguards. The resulting audit trail is therefore part of the research contribution: it records not only the selected final configuration, but also the failed alternatives and the evidence used to reject them. This makes the study easier to evaluate, reproduce, and improve.

For computational communication research, the main implication is that an LLM annotation system can be valuable even when it is not accurate enough for unsupervised final coding. In the present setting, high recall supports screening, candidate retrieval, and the prioritisation of passages for review. Lower precision means that final case-level interpretation and prevalence estimation still require stronger human validation. The most defensible future design would combine a frozen instruction set, stable identifiers, harmonised coder materials, and a genuinely independent validation sample before drawing broader conclusions about outlet-level framing patterns.

Overall, the project achieved its primary technical objective: a functioning, documented, large-scale annotation pipeline and a complete set of production outputs. The next research step is not another unstructured prompt revision. It is a cleaner independent human-validation design, consistent identifiers and codebooks, and targeted work on false positives. With those improvements, the pipeline can support more reliable prevalence estimation, outlet comparison, and extension to additional immigration frames.

Appendices

A Repository and Output Structure

Listing 1: Condensed repository structure

```
economic-framing-annotation/
|-- README.md
|-- config.R
|-- scripts/
|   |-- step0_draw_sample.R
|   |-- step2_1_zeroshot.R
|   |-- step2_2_hardcases.R
|   |-- step3_2_fewshot_cot.R
|   |-- step3_3_hardcases_few.R
|   |-- step5_full_annotation.R
|   |-- step6_report.R
|   |-- step7_* and step8_* human-evaluation scripts
|   +-- utils*.R
|-- data/
|   |-- raw/dataset_10k_translated.csv
|   |-- samples/sample_200.csv and gold_200.csv
|   |-- fewshot/fewshot_training_blocks.csv
|   +-- manual_annotation_1002/
|-- outputs/
|   |-- step2_1/ ... step6/
|   +-- reliability_1002/
|-- reports/
|   |-- prompt_version_log.csv
|   |-- classification_reports/
|   +-- reliability_1002/
|-- artifacts/reporting_1002/
|   |-- tables/
|   |-- figures/
|   +-- narrative/
+-- final_submission_candidate/

economic-framing-annotation/

docs and report workspaces preserve methodology, validation,
security checks, integration logs, and reproducibility notes.

dataset files and completed human-coder files should be handled
according to the applicable data-sharing and privacy rules.
```

B Complete Prompt-Version Log

Table 29: Eight measured prompt-development runs

Run	Step	Main change	T-P	T-R	T-F1	B-P	B-R	B-F1
1	2.1	Baseline v1 instructions	.487	.947	.643	.167	1.000	.286
2	2.1	Logistics/statistics/entitlement exclusions	.556	.789	.652	.750	1.000	.857
3	2.1	Strict NO rules and reminder	.538	.368	.438	.500	1.000	.667
4	2.1	International, criminal, and non-immigration exclusions	.538	.737	.622	.429	1.000	.600
5	3.2	Few-shot, structured explanation, encoding fix	.857	.632	.727	1.000	.333	.500
6	3.2	T5 pressure, B5 degree recognition, B4 examples	.882	.789	.833	.625	1.000	.769
7	3.2	Benefit-negative logistics example	.938	.789	.857	.625	1.000	.769
8	3.2	Benefit-negative legal-status example	.833	.789	.811	.625	1.000	.769

C Pilot Confusion Matrices

Table 30: Economic Threat confusion matrix in the 200-row pilot, Run 8

	Human yes	Human no	Total
LLM yes	15	3	18
LLM no	4	178	182
Total	19	181	200

Table 31: Economic Benefit confusion matrix in the 200-row pilot, Run 8

	Human yes	Human no	Total
LLM yes	5	3	8
LLM no	0	192	192
Total	5	195	200

D Failure and Repair Log

Table 32: Chronological failure and repair log

Date	Step	Failure or observation	Response
22 Apr 2026	2.1	All zero-shot outputs were classified as invalid because the parser expected exact labels.	Developed `clean_label()` v2 with Markdown stripping and regex fallback.
22 Apr 2026	2.2	Repeated annotations revealed unstable Threat and Benefit cases.	Used hard cases to identify vague definitions and missing boundaries.
26 Apr 2026	3	Benefit F1 remained below target on the initial 25-row reference.	Added structured examples and expanded validation.
26 Apr 2026	4	First 10,000-row run produced 4,883 Unknown rows.	Repaired label parsing and reviewed full-corpus distributions.
26 Apr 2026	5	Repaired Phase 1 output had implausibly high and symmetric positive rates.	Redesigned the instructions around T1–T5 and B1–B5.
4 May 2026	3	Threat produced no true positives in a small zero-shot test.	Moved to structured few-shot inference.
4 May 2026	4	Phase 2 still contained 797 Unknown rows and excessive positive rates.	Began precision-focused pilot development.
5 May 2026	Run 3	Blanket strict-NO instruction caused severe Threat recall regression.	Reverted the blanket rule and used targeted exclusions.
5 May 2026	Run 5	Non-ASCII text caused HTTP 400 errors.	Added `clean_text_api()` and row-level retry logic.
5 May 2026	Run 5	Benefit recall collapsed after restrictive examples.	Added explicit positive B4/B5 examples.
7 May 2026	Pilot	Four zero-shot attempts produced zero true positives and undefined F1.	Confirmed that few-shot structured inference was necessary.
7 May 2026	Run 8	Benefit remained below the nominal 0.80 target at F1=.769.	Accepted as a documented exception because recall was 1.000 and further tightening risked missed positives.
Post hoc	Integrity	The development sample script reset row identifiers.	Reconstructed source keys; found 19 overlaps in the submitted 1,002-row sample; reported the limitation.
Post hoc	Reliability	Paired coder files were identical within blocks.	Reported file-based statistics without claiming proven independent agreement.
Post hoc	Codebook	Workbook criterion shorthand differed from final configuration definitions.	Restricted human-versus-LLM analysis to the binary Threat and Benefit labels.

E Final Instruction Excerpts

Economic Threat

Listing 2: Final Threat instruction excerpt

```
Role: You are a trained content analyst. Detect economic
threat framing in one translated German news paragraph.

OCCURRENCE RULE:
Code YES whenever an economic threat frame linked to
immigration is explicitly mentioned, even if it is quoted,
denied, mocked, or refuted.

Code YES if any criterion applies:
T1 economic well-being or prospects are threatened
T2 Germany/EU economic prospects are threatened
T3 labour-market harm
T4 welfare or public-finance strain
T5 costs, burden, overload, capacity limits, or resources

Important NO boundaries:
```

```
- accommodation logistics without cost or burden language
- statistics without explicit shortage or strain
- humanitarian hardship of refugees without receiving-society cost
- general economic news not linked to immigration
- foreign-aid or international-policy stories without German burden
```

Economic Benefit

Listing 3: Final Benefit instruction excerpt

```
Code YES if any criterion applies:
B1 tax revenue, economic growth, GDP, or attracting workers
B2 positive economic effect for Germany or the EU
B3 demographic necessity or pension/workforce contribution
B4 immigration fills a named labour shortage or vacancy
B5 integration or policy creates or prevents loss of an
   explicitly named economic benefit

Important NO boundaries:
- legal permission to work without an economic outcome
- employment assistance without a named shortage filled
- staffing shortages required to manage refugees
- vague optimism or general social contribution
- cultural integration without explicit economic outcome
- macroeconomic news unrelated to immigration
```

F Technical Helper Functions

Listing 4: Simplified robust label parser

```
clean_label <- Vectorize(function(x) {
  if (is.null(x) || is.na(x)) return(NA_character_)
  x <- toupper(trimws(gsub("[\\*`]", "", as.character(x))))
  if (grepl("LABEL:\\s*YES", x)) return("YES")
  if (grepl("LABEL:\\s*NO",  x)) return("NO")
  matches <- regmatches(x, gregexpr("\\b(YES|NO)\\b", x))[[1]]
  if (length(matches) > 0) return(matches[length(matches)])
  NA_character_
})
```

Listing 5: Simplified API text cleaner

```
clean_text_api <- Vectorize(function(x) {
  if (is.null(x) || is.na(x)) return(x)
  x <- iconv(x, to = "UTF-8", sub = "")
  gsub("[^\\x01-\\x7F]", " ", x)
})
```

G Project Timeline

Table 33: Project timeline

Date	Phase	Main activity
22 Apr 2026	Phase 1	Translation of 10,000 rows; initial zero-shot testing; first parser failure; hard-case detection.
26 Apr 2026	Phase 1	Small-reference validation, structured inference, first full production attempt, Unknown-label repair, and over-coding diagnosis.
4 May 2026	Phase 2	Instruction redesign using explicit T1–T5 and B1–B5 criteria; second full-corpus run.
5–7 May 2026	Phase 3	Creation of 200-row pilot pilot reference set; eight measured prompt runs; encoding fix; final prompt selection.
7 May 2026	Production	Final Threat and Benefit annotation of 10,000 paragraphs.
May–July 2026	Evaluation	Human annotation package, submitted coder files, consensus, reliability, LLM comparison, error exports, and report-ready figures.
July 2026	Finalisation	Post-hoc audit, report integration, proofreading, packaging, and documentation.

H Submitted 1,002-Row Confusion Matrices

Table 34: Economic Threat: submitted human consensus versus LLM

	LLM yes	LLM no	Total
Human yes	26	3	29
Human no	43	930	973
Total	69	933	1,002

Table 35: Economic Benefit: submitted human consensus versus LLM

	LLM yes	LLM no	Total
Human yes	14	2	16
Human no	26	960	986
Total	40	962	1,002

I Post-Hoc Integrity Audit

The post-hoc audit reconstructed the original source identifiers of the 200-row development sample using the unique combination `article_id + par_index + group`. The script-created `row_num` values 1–200 did not correspond to the source rows. Nineteen original development rows were found in the submitted 1,002-row sample:

455, 2868, 3788, 4225, 4340, 4555, 5155, 5171, 5566, 5717, 5821, 6034, 6590, 7676, 7778, 9245, 9319, 9836, 9853.

The current results were not recalculated with replacement rows because the submitted human labels correspond to the existing 1,002-row package. The report therefore preserves the submitted arithmetic and clearly states that the evaluation was not a perfectly untouched holdout.

The audit also confirmed that the 10,000-row production output itself was internally complete: 10,000 unique row identifiers, no missing final labels, 731 Threat positives, 416 Benefit positives, and frame-type counts summing to 10,000.

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